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Assignment Title: Agile Sprint Planning & User Story Workshops

Smart E-commerce Order Fulfillment Platform

1. **Identify the Project Vision and Stakeholders**

Project Vision

The vision of the Smart E-commerce Order Fulfillment Platform is to develop an intelligent and automated system that manages the complete order fulfillment process, from order placement to successful delivery. The platform aims to improve operational efficiency by automating inventory management, warehouse allocation, delivery tracking, and customer notifications. It also provides real-time analytics and reporting to help businesses make informed decisions and enhance customer satisfaction.

Project Objectives

- Automate the order fulfillment process.
- Reduce manual work and human errors.
- Improve inventory management. ➢ Optimize warehouse allocation.
- Provide real-time order tracking.
- Increase customer satisfaction.
- Generate business reports for decision-making.

Stakeholders

Stakeholder	Role
Customer	Places orders, tracks deliveries, receives notifications
Administrator	Manages products, users, inventory, and reports
Warehouse Staff	Packs and dispatches customer orders
Delivery Partner	Delivers products and updates delivery status
Business Manager	Monitors reports and business performance
Payment Gateway	Processes online payments

2. **Create Epics and User Stories**

Epic 1: Customer Management

User Story 1

As a customer, I want to register an account so that I can securely purchase products.

User Story 2

As a customer, I want to log in to my account so that I can access my orders and profile.

Epic 2: Product Management

User Story 3

As a customer, I want to search and browse products so that I can find items easily.

Epic 3: Order Management

User Story 4

As a customer, I want to place an order so that I can purchase products online.

User Story 5

As an administrator, I want to manage customer orders so that order processing is efficient.

Epic 4: Inventory Management

User Story 6

As a warehouse staff member, I want to update inventory automatically so that stock information remains accurate.

Epic 5: Delivery Management

User Story 7

As a customer, I want to track my order in real time so that I know when my order will arrive.

Epic 6: Notification System

User Story 8

As a customer, I want to receive notifications about my order status so that I stay informed.

3. **Define Acceptance Criteria**

User Story 1

Given a new customer,

When the customer enters valid registration details,

Then the account should be created successfully and a confirmation message should be displayed.

User Story 4

Given products are available in stock,

When the customer confirms the order,

Then the order should be placed successfully and an order ID should be generated.

User Story 6

Given an order has been confirmed,

When products are dispatched,

Then the inventory quantity should automatically decrease.

User Story 7

Given an order has been shipped,

When the customer checks the tracking page,

Then the current delivery status should be displayed accurately.

User Story 8

Given the order status changes,

When the system updates the order,

Then the customer should receive an email or SMS notification.

4. **Prioritize the Product Backlog**

Priority	**User Story**	**Business Value**
High	User Registration	Essential
High	User Login	Essential
High	Product Search	Essential
High	Place Order	Essential
High	Inventory Update	High
High	Order Management	High
Medium	Delivery Tracking	Medium
Medium	Customer Notifications	Medium
Low	Reports & Analytics	Low
Low	Admin Dashboard Improvements	Low

5. **Plan the Sprint Backlog**

Sprint Backlog (Sprint 1)

Sprint Duration

2 Weeks

Selected User Stories

- User Registration
- User Login
- Product Search
- Place Order

Development Tasks

Task 1

Design Registration Page

Task 2

Develop Login Module

Task 3

Create Product Database

Task 4

Develop Product Search

Task 5

Implement Shopping Cart

Task 6

Develop Order Placement Module

Task 7

Perform Unit Testing

Task 8

Fix Bugs

6. **Estimate Effort Using Story Points**

7. User Story	Story Points	Justification
User Registration	3	Simple form with validation
User Login	2	Basic authentication
Product Search	5	Search and filtering functionality
Place Order	8	Includes cart, payment, and order confirmation
Inventory Update	5	Database synchronization
Delivery Tracking	5	Real-time tracking integration
Notifications	3	Email/SMS integration

Justification

The Fibonacci sequence (1, 2, 3, 5, 8, 13…) is used because it helps estimate development effort based on complexity and uncertainty. Simpler tasks such as login require fewer story points, while more complex features like order placement involve multiple components and therefore require higher story points.

8. **Present the Sprint Goal and Expected Deliverables**

Sprint Goal

Sprint Goal

The primary goal of Sprint 1 is to build the essential customer-facing features of the Smart E-commerce Order Fulfillment Platform. During this sprint, the development team focuses on creating a secure and user-friendly system that allows customers to create an account, log in, browse available products, view product details, add items to a shopping cart, and place orders successfully. These features represent the core functionality of any e-commerce application and ensure that customers can complete the shopping process without difficulties.

In addition to implementing these features, the sprint also aims to establish the basic project architecture, database connectivity, and user interface design that will support future development. Security measures such as password encryption, user authentication, and input validation are included to ensure safe user access. The team will also verify that all implemented features work correctly through testing and bug fixing.

By the end of Sprint 1, the platform should provide a functional customer module that serves as the foundation for integrating advanced features in later sprints, including inventory management, payment gateway integration, delivery tracking, order history, notifications, and administrative controls.

Expected Deliverables

At the completion of **Sprint 1**, the following deliverables are expected:

1. **User Registration Module**
   - Customers can create a new account by providing personal details such as name, email, phone number, and password.

- ➢ Input validation ensures accurate data entry.
- ➢ Passwords are securely stored using encryption.

2. **User Login and Authentication**
    - ➢ Registered users can log in securely using their credentials.
    - ➢ Authentication mechanisms prevent unauthorized access.
    - ➢ Session management allows users to remain logged in during their shopping session.
3. **Product Browsing Module**
    - ➢ Customers can view a list of available products.
    - ➢ Product information includes images, descriptions, prices, and availability.
    - ➢ Basic search and category filtering features improve product discovery.
4. **Shopping Cart Functionality**
    - ➢ Users can add, update, or remove products from the shopping cart.
    - ➢ The cart automatically calculates total cost and item quantity.
5. **Order Placement Module**
    - ➢ Customers can review their selected products and place an order successfully.
    - ➢ The system records order details in the database.
    - ➢ Users receive an order confirmation message after successful submission.
6. **Database Design and Integration**
    - ➢ Creation of essential database tables for users, products, shopping cart, and orders.
    - ➢ Proper relationships between tables are established to maintain data consistency.
7. **Responsive User Interface**

- Basic web pages are designed using a responsive layout.

8. **Testing and Bug Fixes**
   - Functional testing of all implemented features.
   - Validation of user inputs.
   - Identification and correction of software defects before sprint completion.
9. **Sprint Documentation**
   - Updated product backlog and sprint backlog.
   - Source code repository with version control.
   - Sprint review demonstration.
   - Sprint retrospective documenting lessons learned and improvement plans.

Sprint Outcome

At the end of Sprint 1, the Smart E-commerce Order Fulfillment Platform will provide a working customer module where users can register, log in, browse products, manage their shopping cart, and place orders. This working prototype forms the core of the application and provides a stable foundation for future sprints that will introduce features such as inventory management, payment processing, order tracking, delivery management, customer notifications, analytics, and administrative functionalities.